AIonOS

AIonOS × Portek

Autonomous Port Network Intelligence

Transform fragmented terminal operations into a real-time, predictive, cross-port control layer — without replacing core systems.

5-Intelligent Decision Systems

Prepared for Portek leadership

THE BIG IDEA

Executive thesis: make every Portek terminal visible, comparable and self-optimising

Portek's value is constrained less by terminal capability, and more by fragmented operating intelligence across ports.

Control Tower

Single live view of vessels, berths, yard, shipments and equipment across regions.

Decision latency ↓ 30–50%

Agentic Twin

Simulate berth, crane, yard and dispatch decisions before congestion happens.

Dwell time ↓ 20–25%

Asset Intelligence

Predict crane and equipment risk, standardise maintenance and reduce unplanned downtime.

Availability ↑ 10–15%

Transformation target

From "manual reporting by terminal" to an AI-powered port operating layer that senses, predicts and recommends / executes actions.

1	source of truth
4	priority use cases
90d	pilot proof point
18m	autonomous scale

Portek's operating reality: strong port know-how, but distributed data gravity

The opportunity is not "new IT"; it is cross-port operating leverage.

Global multi-terminal footprint

Singapore HQ; terminals / projects across Indonesia, Africa, Europe and Vietnam.

Asset-intensive operations

Cranes, yards, berths and handling equipment drive throughput and cost.

Semi-independent local systems

Vessel, yard, shipment and engineering data remain dispersed by location.

HQ visibility gap

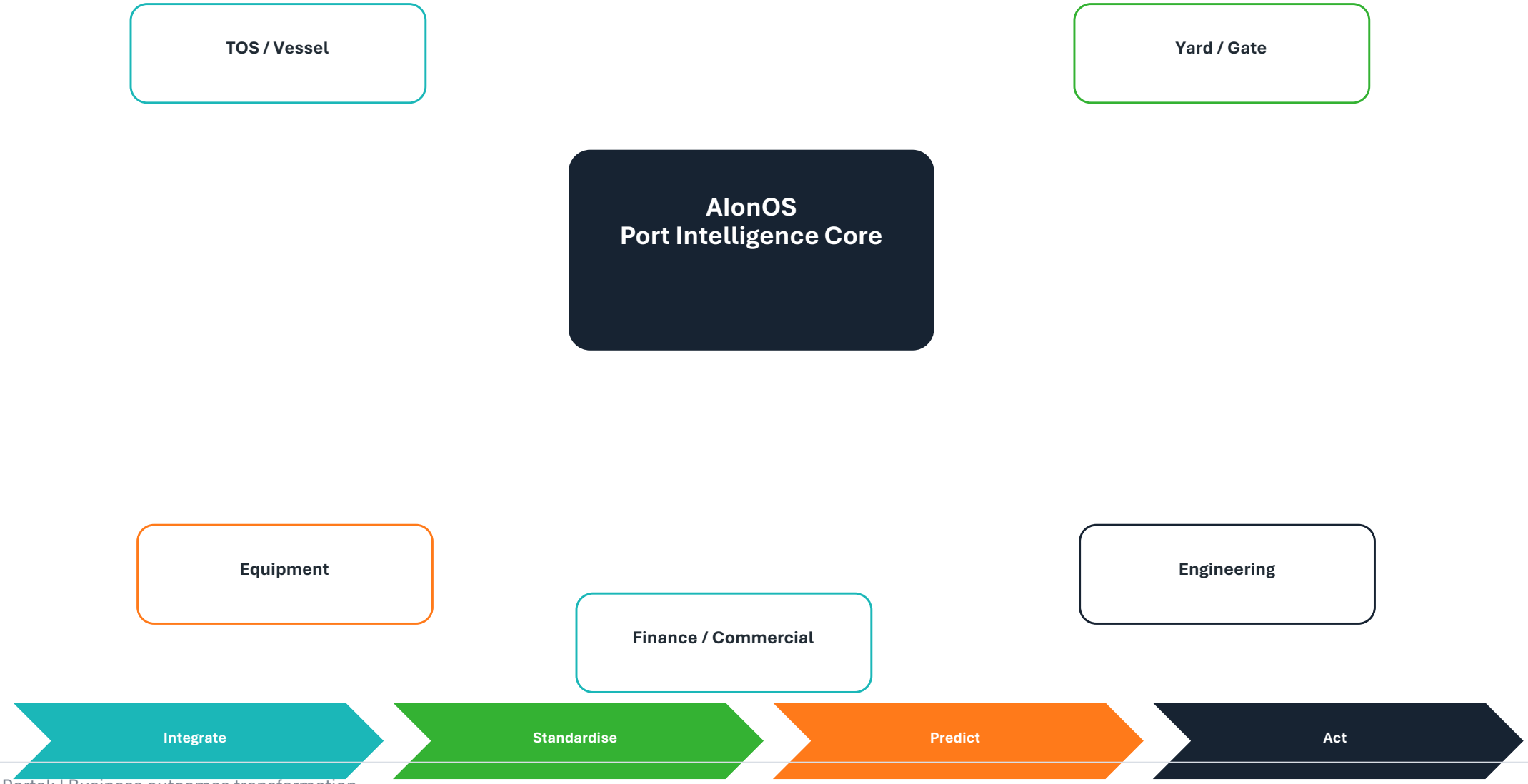
Manual reconciliation slows decisions and prevents cross-port benchmarking.

Pain Point: no unified, real-time shipment / vessel / asset view across ports.

TARGET STATE

AIonOS transformation blueprint: a port intelligence layer above existing systems

Keep Portek's local systems. Add a governed intelligence layer that standardises data, predicts risk and orchestrates action.



Four use cases Portek can pilot fast — and scale across regions

01 Cross-port visibility cockpit

Vessels, berths, yard inventory, equipment availability.

Reporting effort ↓ 50–70%

02 Berth + crane scheduling twin

Predict congestion and reallocate berth windows / crane gangs.

Crane productivity ↑ 10–15%

03 Yard + dispatch orchestrator

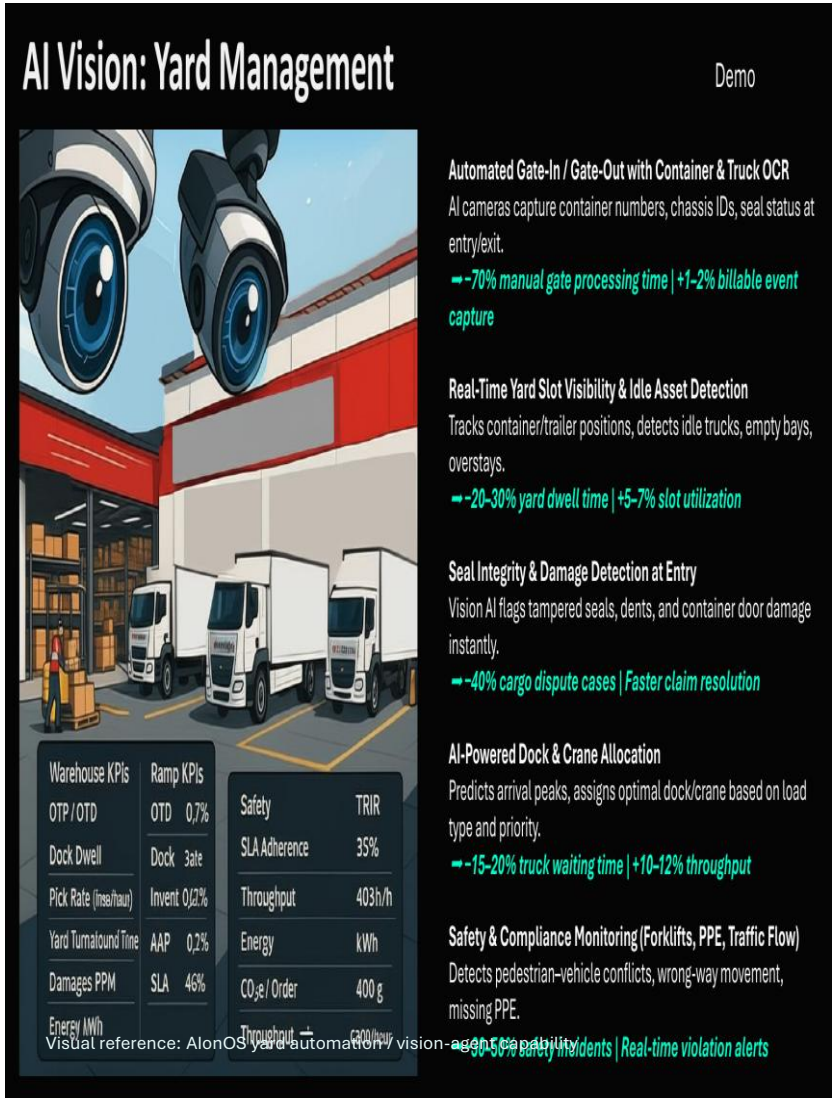
Resequence trucks / rail loads against live ETAs and customs cut-offs.

Port dwell ↓ 20–25%

04 Equipment reliability AI

Predict crane / handling equipment failure and plan parts / maintenance.

Unplanned downtime ↓ 15–20%



Visual reference: AIonOS yard automation / vision-agent capability

Reference architecture: integrate once, operate many terminals

1. Connect	MCP adapters for TOS, ERP, yard, vessel, engineering and spreadsheets.	Connector live < 2 weeks
2. Govern	Unified data model, RBAC, audit trails and access controls.	100% data access audited
3. Sense	Real-time streams and data products for agents and analytics.	Data freshness < 5 min
4. Act	Agentic workflows for berth, yard, dispatch and maintenance.	Response latency < 120 sec

Outcome: one Portek data foundation for HQ visibility and terminal-level autonomy.

Value at stake: measurable business outcomes, not “AI experiments”

Target KPIs should be baselined in the 90-day pilot, then managed as a global operating scorecard.

Decision speed	Manual reports → live insights	**30–50% faster HQ decisions**
Throughput	Berth / crane / yard constraints	**10–15% productivity uplift**
Dwell & SLA	Waiting, rehandles, cut-off misses	**20–25% dwell reduction**
Asset uptime	Crane / equipment failures	**15–20% downtime reduction**
Commercial control	Fragmented charges / leakage	**1–2% billable event capture**

Management lens: one dashboard, four controllable levers — visibility, throughput, uptime, cash.

Roadmap: start narrow, prove ROI, then industrialise across ports

0–4 weeks

Discover + baseline

Map systems, KPIs, data fields and 1–2 pilot terminals.

Validated data map + pilot KPI baseline

5–12 weeks

Control tower MVP

Unified vessel / shipment / equipment dashboard with GenBI interface.

Live cross-port cockpit + exception triage

3–9 months

Agentic optimisation

Berth, crane, yard and dispatch twins with human-in-loop actions.

Dwell, productivity and uptime improvements

9–18 months

Autonomous scale

Replication playbook across regions, best-practice benchmarking and agents.

Portek operating intelligence layer

Principle: no rip-and-replace. Build an AI orchestration layer that keeps local terminal context intact.

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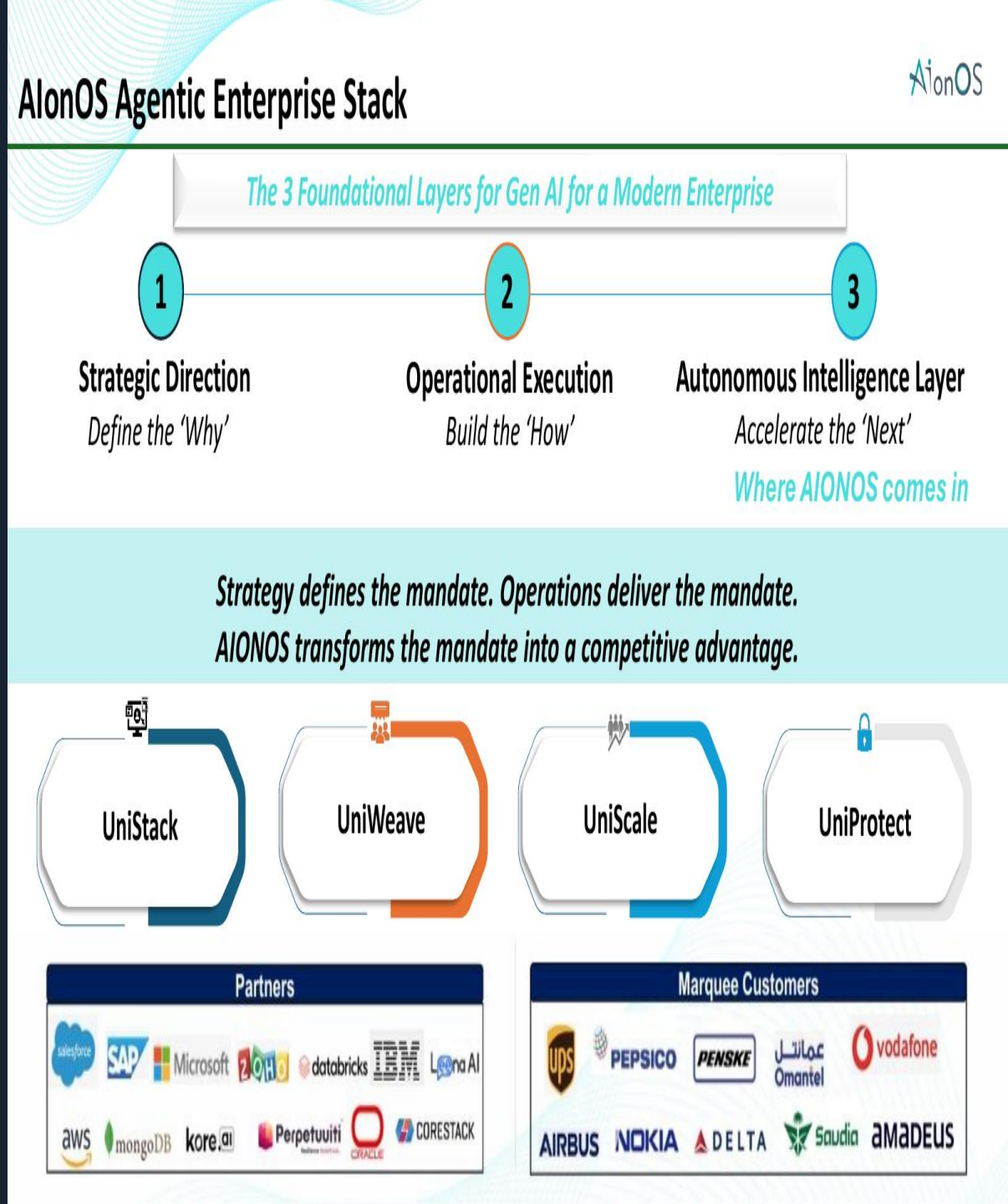
Recommended next move

Launch a 90-day Portek Control Tower + Agentic Twin pilot across 1–2 terminals.

Decision	Select pilot terminal(s) + executive sponsor
Data	Confirm TOS / yard / vessel / engineering sources
KPI	Baseline dwell, crane productivity, equipment uptime, reporting effort
Build	Deploy MVP cockpit + exception playbooks

Outcome: Portek becomes a learning port network — every terminal improves every other terminal.